Exercise 10

On scalar product of two vectors

From the school book | Understand | Apply | Higher Order Thinking Skills

1 If $\vec{A}$ and $\vec{B}$ are two vectors and the measure of the angle between them is $135°$,
$\|\vec{A}\| = 6\sqrt{2}$, $\|\vec{B}\| = 10\sqrt{2}$ Find $\vec{A} \cdot \vec{B}$ « $-60\sqrt{2}$ »

2 **Find $\vec{A} \cdot \vec{B}$ in each of the following cases :**

(1) $\vec{A} = (5 , 1 , -2)$, $\vec{B} = (4 , -4 , 3)$

(2) $\vec{A} = (-2 , 4 , 2)$, $\vec{B} = 5\hat{i} - 3\hat{j} + 6\hat{k}$

(3) $\vec{A} = -3\hat{i} - 2\hat{j} - \hat{k}$, $\vec{B} = 6\hat{i} + 4\hat{j} + 2\hat{k}$

(4) $\vec{A} = \hat{j}$, $\vec{B} = 2\hat{i} - 3\hat{k}$

3 **Find the measure of the angle between the two vectors $\vec{A}$, $\vec{B}$ if :**

(1) $\|\vec{A}\| = 16$, $\|\vec{B}\| = 10$, $\vec{A} \cdot \vec{B} = 80$ « $60°$ »

(2) $\|\vec{A}\| = 25$, $\|\vec{B}\| = 18$, $\vec{A} \cdot \vec{B} = -225\sqrt{2}$ « $135°$ »

(3) $\|\vec{A}\| = 6$, $\|\vec{B}\| = \frac{5}{2}\|\vec{A}\|$, $\vec{B} \cdot \vec{A} = 81$ « $25° 51\grave{}$ »

(4) $\|\vec{A}\| = 32$, $\|\vec{B}\| = 25$, $\vec{B} \cdot \vec{A} = -480$ « $126° 52\grave{}$ »

4 **Find the measure of the angle between the two vectors in each of the following cases :**

(1) $(5 , 1 , -2)$, $(1 , 1 , -1)$

(2) $(4 , 1 , 2)$, $(0 , -2 , 1)$

(3) $4\hat{i} + 3\hat{j} + 7\hat{k}$, $2\hat{i} + 5\hat{j} + 4\hat{k}$

(4) $-3\hat{j} + 8\hat{k}$, $10\hat{i} - 5\hat{k}$

5 **If $\hat{i}$, $\hat{j}$, $\hat{k}$ are unit vectors of right-hand system find each of the following :**

$\hat{i} \cdot \hat{i}$, $\hat{j} \cdot \hat{j}$, $\hat{k} \cdot \hat{k}$, $\hat{i} \cdot \hat{j}$, $(3\hat{i}) \cdot (4\hat{k})$, $(5\hat{i}) \cdot (2\hat{i})$

6 **Choose the correct answer from those given :**

(1) *(2nd Session 2017)* If $\vec{A} = 2\hat{i} + 3j - \hat{k}$, $\vec{B} = 4\hat{i} - \hat{j}$, then $\vec{A} \cdot \vec{B} =$ .....

(a) 5 (b) 4 (c) 3 (d) 2

(2) If $\|\vec{A}\| = 5$, $\vec{B} = (-1, 2, 2)$ and the measure of the angle between the two vectors is $30°$, then $\vec{A} \cdot \vec{B}$ = ...

(a) $(-5, 10, 10)$ (b) $7.5\sqrt{3}$ (c) 7.5 (d) -7.5

(3) If $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{B} = 3\hat{i} + 2\hat{j} - \hat{k}$, then $(\vec{A} + 3\vec{B}) \cdot (2\vec{A} - \vec{B})$ = ...

(a) zero (b) 4 (c) 5 (d) 6

(4) If A $(2, 1, 3)$, B $(3, 5, -2)$, C $(-1, 4, 0)$, then $\overrightarrow{AB} \cdot \overrightarrow{AC}$ = ...

(a) 5 (b) 24 (c) 17 (d) 13

(5) If $2\vec{A} = -6\hat{i} - 4\hat{j}$, $3\vec{B} = 9\hat{i} - 15\hat{k}$, then $3\vec{A} \cdot 2\vec{B}$ = ...

(a) -54 (b) zero (c) 6 (d) 54

(6) $\vec{A}$, $\vec{B}$ are two unit vectors enclosing angle of measure $45°$, then $\vec{A} \cdot \vec{B}$ = ...

(a) $45°$ (b) $\sqrt{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $-\sqrt{2}$

(7) If $\|\vec{A}\| = 15$, $\|\vec{B}\| = 20$, $\vec{A} \cdot \vec{B} = -150\sqrt{3}$, then measure of the angle between the two vectors = ...

(a) $30°$ (b) $120°$ (c) $135°$ (d) $150°$

(8) If θ is the included angle between the two vectors $\vec{A} = (2, 0, 2)$, $\vec{B} = (0, 0, 4)$, then θ = ...

(a) $30°$ (b) $45°$ (c) $60°$ (d) $90°$

(9) If θ is the angle between the two vectors $\vec{A} = (-2, -6, 1)$ and $\vec{B} = (2, 6, -1)$, then θ = ...

(a) $30°$ (b) $60°$ (c) $120°$ (d) $180°$

(10) Measure of the angle between the two vectors $\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + \hat{k}$ is .....

(a) $\cos^{-1}\left(\frac{1}{\sqrt{15}}\right)$ (b) $\cos^{-1}\left(\frac{4}{\sqrt{15}}\right)$ (c) $\cos^{-1}\left(\frac{4}{15}\right)$ (d) $\frac{\pi}{2}$

(11) Cosine the angle between the two vectors $\vec{A} = (1, -3, 0)$, $\vec{B} = (2, 0, 1)$ equals .....

(a) $\frac{\sqrt{2}}{5}$ (b) $5\sqrt{2}$ (c) $\frac{5}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}$

(12) The measure of the angle between the two vectors $3\hat{i} - \hat{j}$, $-4\hat{i} + 6\hat{j}$ equals .....

(a) $35° 40'$ (b) $108° 42'$ (c) $37° 52' 30''$ (d) $142° 7' 30''$

(13) If $\|\vec{A}\| = 4$, $\|\vec{B}\| = 6$ and the measure of the angle between $\vec{A}$ and $\vec{B}$ equals $60°$, then $(3\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B})$ = ...

(a) 48 (b) 36 (c) 12 (d) -12

(14) If $\vec{A}$, $\vec{B}$ are non-zero vectors and $\vec{A} \cdot \vec{B} = 0$, then the two vectors .....

(a) are parallel. (b) lie in the $\mathcal{X}$ y-plane.

(c) are perpendicular. (d) have same norm.

(15) The two vectors $\vec{A} = (1 , 3 , -2)$, $\vec{B} = (-2 , -6 , 4)$ are .....

(a) parallel and have the same direction.

(b) parallel but in opposite directions.

(c) perpendicular.

(d) the measure of the angle between them is 60°

(16) If $\vec{A}$, $\vec{B}$ are two perpendicular unit vectors , then $(\vec{A} - 2\vec{B}) \cdot (3\vec{A} + 5\vec{B})$ = ...

(a) -8 (b) -7 (c) 24 (d) 0

(17) If $\vec{A} = 3\hat{i} + 3\hat{j} + m\hat{k}$, $\vec{B} = -6\hat{i} - 4\hat{j} + 6\hat{k}$ and $\vec{A} \perp \vec{B}$, then m = ...

(a) -4 (b) 5 (c) 7 (d) 24

(18) If $\vec{A} = (k , -3 , 1)$, $\vec{B} = (2 , 3 , -k)$ are perpendicular , then the value of k = ...

(a) -9 (b) -3 (c) 9 (d) 18

(19) If $\vec{A} \cdot \vec{B} = 4$ and $\vec{A} = m\hat{i} + \hat{j} + 4\hat{k}$, $\vec{B} = 2\hat{i} + 6\hat{j} + 3\hat{k}$, then m = ...

(a) 4 (b) 7 (c) -4 (d) -7

(20) If $(3\vec{A} \cdot 2\vec{B}) = 12$, then $(-\vec{B} \cdot 5\vec{A})$ = ...

(a) -10 (b) -5 (c) 5 (d) 10

(21) If $\vec{A} \cdot \vec{B} = 13$, then the angle between the two vectors $\vec{A}$ and $\vec{B}$ is .....

(a) acute (b) right (c) obtuse (d) straight

(22) If $\vec{A} \perp \vec{B}$ and $[(2\vec{A} - 4\vec{C}) \cdot 3\vec{B}] = 108$, then $\vec{B} \cdot \vec{C}$ = ...

(a) -7 (b) -8 (c) -9 (d) -12

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7 ABCD is a square with side length 10 cm. , **find :**

(1) $\overrightarrow{AB} \cdot \overrightarrow{DC}$ **(2)** $\overrightarrow{AB} \cdot \overrightarrow{BC}$ **(3)** $\overrightarrow{AB} \cdot \overrightarrow{CA}$ « 100 , zero , –100 »

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8 ABC is an equilateral triangle of side length 12 cm. , **find :**

(1) $\overrightarrow{AB} \cdot \overrightarrow{AC}$ **(2)** $\overrightarrow{AC} \cdot \overrightarrow{CB}$ **(3)** $(4\overrightarrow{AB}) \cdot (3\overrightarrow{CB})$ « 72 , – 72 , 864 »

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9 *(1st Session 2017)* ABCD is rectangle in which AB = 6 cm. , BC = 8 cm. , **find :**

(1) $\overrightarrow{AB} \cdot \overrightarrow{AC}$ **(2)** $\overrightarrow{AB} \cdot \overrightarrow{CD}$

(3) The component of $\overrightarrow{CD}$ in the direction of $\overrightarrow{BC}$ « 36 , – 36 , 0 »

10 ABCD is a rectangle , AB = 6 cm. , BC = 8 cm. **Find :**

(1) $3\overrightarrow{AD} \cdot 2\overrightarrow{CD}$ | (2) $(-2\overrightarrow{AB}) \cdot (\overrightarrow{AC})$

(3) $\overrightarrow{BC} \cdot \left(\frac{5}{2}\overrightarrow{DB}\right)$ | (4) $\overrightarrow{AC} \cdot \overrightarrow{BD}$

« 0 , – 72 , – 160 , 28 »

11 ABCD is a parallelogram , m (∠ BAC) = 90° , BC = 2 AB = 8 cm. **Find :**

(1) $\overrightarrow{CA} \cdot \overrightarrow{CB}$ | (2) $\left(\frac{1}{3}\overrightarrow{AD}\right) \cdot \left(\frac{1}{2}\overrightarrow{DC}\right)$

(3) $\overrightarrow{AB} \cdot \left(\sqrt{3}\overrightarrow{AD}\right)$ | (4) $(7\overrightarrow{BD}) \cdot (\overrightarrow{AC})$

« 48 , $\frac{-8}{3}$, $-16\sqrt{3}$, 336 »

12 ABCD is a rhombus , the length of its diagonals $\overrightarrow{AC}$, $\overrightarrow{BD}$ are 24 , 32 cm. **Find :**

(1) $\overrightarrow{AB} \cdot (-2\overrightarrow{DC})$ (2) $\left(\frac{1}{2}\overrightarrow{AC}\right) \cdot \overrightarrow{DB}$ (3) $\overrightarrow{AD} \cdot \overrightarrow{DB}$ (4) $\overrightarrow{BA} \cdot \overrightarrow{BC}$

« – 800 , 0 , – 512 , 112 »

13 ABC is a right-angled triangle at ∠ B , AB = 5 cm. , BC = 12 cm. D is the midpoint of $\overline{AC}$ **Find :**

(1) $\overrightarrow{AB} \cdot \overrightarrow{AC}$ (2) $\overrightarrow{BC} \cdot \left(\frac{1}{2}\overrightarrow{BA}\right)$ (3) $\overrightarrow{BA} \cdot \overrightarrow{BD}$ (4) $\overrightarrow{DB} \cdot \left(\frac{1}{4}\overrightarrow{BC}\right)$

« 25 , 0 , $12\frac{1}{2}$, – 18 »

14 ABCD is a trapezium $\overline{AD} \parallel \overline{BC}$, m (∠ A) = m (∠ B) = 90° , AD = $\frac{1}{2}$ BC = 24 cm. , CD = 26 cm. **Find :**

(1) $\overrightarrow{CD} \cdot \overrightarrow{CB}$ (2) $\overrightarrow{DB} \cdot \overrightarrow{BC}$ (3) $\overrightarrow{DC} \cdot \overrightarrow{AB}$ « 1152 , – 1152 , 100 »

15 ABCDEF is a regular hexagon , of side length 8 cm. **Find :**

(1) $(\overrightarrow{CA} + \overrightarrow{AF}) \cdot \overrightarrow{AD}$ (2) $(\overrightarrow{AB} - \overrightarrow{EF}) \cdot \overrightarrow{AD}$ « –128 , 192 »

16 If $\overrightarrow{A} = 4\hat{i} - 3\hat{j}$, $\overrightarrow{B} = \hat{i} + 2\hat{j} - \hat{k}$, $\overrightarrow{C} = -\hat{i} + \hat{j} + 2\hat{k}$ **Find :**

$2\overrightarrow{C} - \left[\frac{1}{5}\overrightarrow{A} \cdot [(2\overrightarrow{B}) + (3\overrightarrow{C})]\right]\overrightarrow{B}$

17 Prove that : Δ ABC is right-angled triangle where A (2 , – 1 , 1) , B (1 , – 3 , – 5) , C (3 , – 4 , – 4)

18 If $\overrightarrow{A} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\overrightarrow{B} = 3\hat{i} - \hat{j} + 2\hat{k}$

Prove that the vector $(\overrightarrow{A} + \overrightarrow{B})$ is perpendicular to the vector $(\overrightarrow{A} - \overrightarrow{B})$

19 **Choose the correct answer from those given :**

(1) If $\overrightarrow{A}$, $\overrightarrow{B}$ are perpendicular , $\overrightarrow{A}$ = (5 , – 4 , 0) , then $\overrightarrow{B}$ could be .....

(a) (3 , 4 , 3) (b) (8 , 10 , – 7) (c) (1 , 1 , 5) (d) (0 , 1 , 0)

(2) If $\overrightarrow{A}$ = (– 1 , 2 , 2) , $\overrightarrow{B}$ is a unit vector then $\overrightarrow{A} \cdot \overrightarrow{B}$ could be equal to .....

(a) 7 (b) 4 (c) 2 (d) – 4

(3) $\vec{A}$, $\vec{B}$ are two unit vectors , their scalar product is $\left(-\frac{1}{2}\right)$, then measure of the angle between them is .....

(a) 30° (b) 60° (c) 120° (d) 150°

(4) If $\vec{A}$, $\vec{B}$ are two unit vectors , then $\vec{A} \cdot \vec{B} \in$ .....

(a) $]0 , 1[$ (b) $]-1 , 1[$ (c) $[-1 , 1]$ (d) $\mathbb{R}^+$

(5) $\vec{A}$ and $\vec{B}$ are two unit vectors , θ is the measure of the included angle between them , then $\vec{A} + \vec{B}$ is a unit vector if .....

(a) $\theta = \frac{\pi}{3}$ (b) $\theta = \frac{\pi}{2}$ (c) $\theta = \frac{2\pi}{3}$ (d) $\theta = \pi$

(6) The component of vector $\vec{A}$ in direction of $\vec{B}$ where θ is the included angle between them = ...

(a) $(\vec{A} \cdot \vec{B}) \cos\theta$ (b) $\|\vec{B}\| \|\vec{A}\|$ (c) $\|\vec{A}\| \cos\theta$ (d) $\|\vec{A}\| \|\vec{B}\| \cos\theta$

(7) If $\|\vec{A}\| = 4$, $\|\vec{B}\| = 6$, the measure of the angle between the two vectors is 30° , then the component of vector $\vec{B}$ in direction of $\vec{A}$ is .....

(a) $12\sqrt{3}$ (b) $9\sqrt{3}$ (c) $3\sqrt{3}$ (d) $2\sqrt{3}$

(8) If $\vec{A} = (-1 , 4 , 2)$, $\vec{B} = (2 , 2 , 1)$, then the component of $\vec{A}$ in direction of $\vec{B}$ = ...

(a) $\frac{4}{3}$ (b) $\frac{-8}{3}$ (c) $\frac{8}{3}$ (d) 5

(9) If $\vec{A} = 4\hat{i} - 3\hat{j} + 5\hat{k}$, then the component of $\vec{A}$ in direction of z-axis equals .....

(a) 4 (b) 3 (c) – 3 (d) 5

(10) The projection of vector of $\vec{A} = (2 , 3 , -1)$ in direction of vector of $\vec{B} = 3\hat{i} + 4\hat{j}$ equals .....

(a) 18 (b) $\frac{18}{5}$ (c) $\frac{-18}{5}$ (d) $\frac{18}{25}$

(11) The algebraic component of the force $\vec{F} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ in the direction of $\overrightarrow{AB}$ where A (1 , 4 , 0) and B (– 1 , 2 , 3) is .....

(a) 21 (b) 17 (c) $\sqrt{13}$ (d) $\sqrt{17}$

(12) If the vector $\vec{E} = \vec{A} + \vec{B}$ is perpendicular to the vector $\vec{F} = \vec{A} - \vec{B}$, then .....

(a) $\vec{A} \parallel \vec{B}$ (b) $\vec{A} \perp \vec{B}$ (c) $\vec{B} = \vec{O}$ (d) $\|\vec{A}\| = \|\vec{B}\|$

(13) ABC is an isosceles triangle , $AB = AC = 6\sqrt{3}$ cm. , $m(\angle A) = 120°$, then $\overrightarrow{BA} \cdot \overrightarrow{BC}$ = ...

(a) $54\sqrt{3}$ (b) 108 (c) 81 (d) 162

(14) ABCD is a square of side length 10 cm. , then $\overrightarrow{AB} \cdot \overrightarrow{CA}$ = ...

(a) 100 (b) $100\sqrt{2}$ (c) -100 (d) $-100\sqrt{2}$

(15) If $\overrightarrow{A} = 3\hat{i} - 4\hat{j}$, $\overrightarrow{B} = 12\hat{i} + 5\hat{j}$, θ is the angle between the two vectors $\overrightarrow{A}$, $\overrightarrow{B}$, then $\tan\theta$ = ...

(a) $\frac{16}{65}$ (b) $\frac{63}{65}$ (c) $\frac{63}{16}$ (d) $\frac{16}{63}$

(16) If $\overrightarrow{A} = (1 , -2 , 1)$, $\overrightarrow{B} = (-2 , 1 , 2)$, then the vector component of $\overrightarrow{A}$ in the direction of $\overrightarrow{B}$ = ...

(a) $\left(\frac{4}{9} , \frac{-2}{9} , \frac{-4}{9}\right)$ (b) $\left(\frac{4}{9} , \frac{2}{9} , \frac{4}{9}\right)$ (c) $\left(\frac{-4}{9} , \frac{-2}{9} , \frac{-2}{9}\right)$ (d) $\left(\frac{4}{9} , \frac{2}{9} , \frac{-4}{9}\right)$

(17) ABC is a right-angled triangle at $\angle B$, AB = 6 cm. , BC = 8 cm. , D is the midpoint of $\overline{AC}$, then the algebraic projection of vector $\overrightarrow{BD}$ in direction of $\overrightarrow{AB}$ = ...

(a) 3 (b) -3 (c) 4 (d) -4

(18) If $\overrightarrow{A} + \overrightarrow{B} + \overrightarrow{C} = \overrightarrow{0}$ and $\|\overrightarrow{A}\| = 2$, then $\overrightarrow{A} \cdot \overrightarrow{B} + \overrightarrow{A} \cdot \overrightarrow{C}$ = ...

(a) 8 (b) 4 (c) -4 (d) -8

(19) If $\overrightarrow{A}$, $\overrightarrow{B}$ are non-zero vectors and $\overrightarrow{A} \cdot \overrightarrow{B} = 0$, then the component of $\overrightarrow{A}$ in direction of $\overrightarrow{B}$ equals .....

(a) $\|\overrightarrow{A}\|$ (b) $\|\overrightarrow{B}\|$ (c) 1 (d) zero

(20) If $\|\overrightarrow{A}\| = 6$, $\|\overrightarrow{B}\| = 4$ and the component of $\overrightarrow{A}$ in direction of $\overrightarrow{B}$ equals 3 , then the component of $\overrightarrow{B}$ in direction of $\overrightarrow{A}$ equals .....

(a) -2 (b) 2 (c) -8 (d) 8

(21) If $\left(\overrightarrow{A} \cdot \left(\overrightarrow{B} - \overrightarrow{C}\right)\right)$ = zero , $\overrightarrow{A} = 2\overrightarrow{B}$, $\|\overrightarrow{C}\| = 2\|\overrightarrow{B}\|$, then measure of the angle between $\overrightarrow{A}$ and $\overrightarrow{C}$ = ...

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

(22) If $\|\overrightarrow{A}\| = 2$, $\|\overrightarrow{B}\| = 3$, $\cos\theta = \frac{2}{3}$ where θ is the measure of the angle between $\overrightarrow{A}$ and $\overrightarrow{B}$, then $\|2\overrightarrow{A} - 3\overrightarrow{B}\|$ = ...

(a) 8 (b) 7 (c) 6 (d) 5

(23) If $\|\overrightarrow{A}\| = 2$, $\|\overrightarrow{B}\| = 3$, $\|\overrightarrow{A} - \overrightarrow{B}\| = \sqrt{19}$, then the measure of the angle between $\overrightarrow{A}$ and $\overrightarrow{B}$ = ...

(a) 30° (b) 60° (c) 90° (d) 120°

(24) If $\vec{A} \perp \vec{B}$ and $\| \vec{A} + \vec{B} \| = 12$, then $\| \vec{A} - \vec{B} \| =$ .....

(a) 6 (b) 12 (c) $12\sqrt{2}$ (d) $32\sqrt{3}$

(25) *(Trial 2021)* If $\vec{A}$, $\vec{B}$ are two vectors and $\| \vec{A} \| = 5$, the component of $\vec{B}$ in direction of $\vec{A}$ is 3 , then $\vec{A} \cdot \vec{B} =$ .....

(a) 15 (b) $\frac{5}{3}$ (c) $\frac{3}{5}$ (d) 8

(26) If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are three non-zero vectors and $\vec{A} + \vec{B} = \vec{C}$ and $\| \vec{A} \| = \| \vec{B} \|$, $\| \vec{C} \| = \sqrt{3} \| \vec{A} \|$, then measure of the angle between $\vec{B}$ and $\vec{C}$ equals .....

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

(27) If $\vec{A}$, $\vec{B}$ and $2\vec{A} + \vec{B}$ are non-zero vectors and $\| \vec{A} + \vec{B} \| = \| \vec{A} \|$, then $2\vec{A} + \vec{B}$, $\vec{B}$ are .....

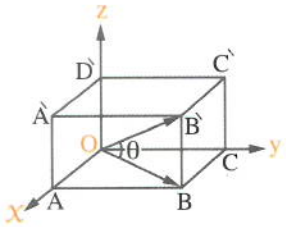
(a) parallel. (b) perpendicular.

(c) including an angle of measure 30° (d) including an angle of measure 60°

(28) In the opposite figure :

ABCOA`B`C`D` is a cuboid

, B` (3 , 5 , 4) , then $\theta \approx$ ..... (to the nearest degree)



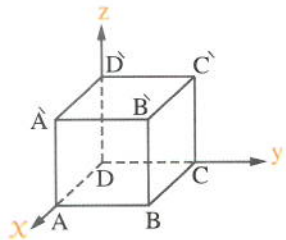
(a) 27° (b) 34°

(c) 48° (d) 64°

(29) In the opposite figure :

ABCDA`B`C`D` is a cube of side length unity

, then $\overrightarrow{AB} \cdot \overrightarrow{BD} =$ .....



(a) – 1 (b) zero

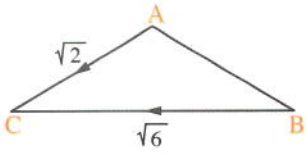
(c) 1 (d) $\frac{1}{2}$

(30) In the opposite figure :

If $\| \overrightarrow{BC} \| = \sqrt{6}$, $\| \overrightarrow{AC} \| = \sqrt{2}$

, $\overrightarrow{BA} = (-1 , 0 , 1)$

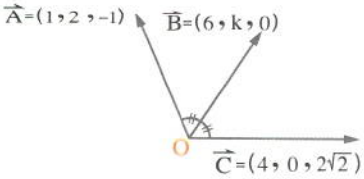
, then $\overrightarrow{BA} \cdot \overrightarrow{BC} =$ .....



(a) $\sqrt{3}$ (b) – 3 (c) 3 (d) $\frac{\sqrt{3}}{2}$

(31) **In the opposite figure :**

The value of k = ...

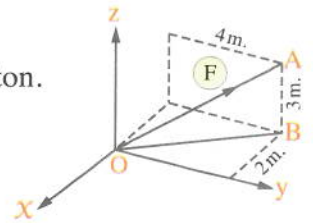


(a) – 3 (b) 3

(c) $\sqrt{3}$ (d) 2

(32) **In the opposite figure :**

The component of the force $\vec{F}$ whose magnitude $12\sqrt{29}$ newtons and acts on $\overrightarrow{OA}$ in direction of $\overrightarrow{OB}$ is ..... newton.



(a) $24\sqrt{29}$

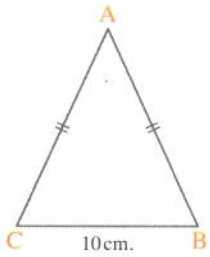
(b) $24\sqrt{5}$

(c) $-48\sqrt{29}$

(d) $6\sqrt{29}$

(33) **In the opposite figure :**

$\overrightarrow{CA} \cdot \overrightarrow{CB}$ = ...

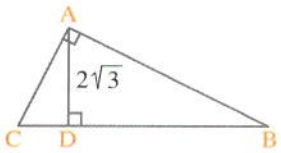


(a) 100 (b) 75

(c) 50 (d) 25

(34) **In the opposite figure :**

$\overrightarrow{AD} \cdot (\overrightarrow{AB} + \overrightarrow{AC})$ = ...

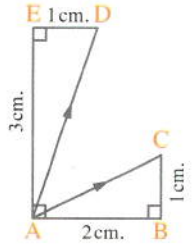


(a) 36 (b) 24

(c) 12 (d) 6

(35) **In the opposite figure :**

$\overrightarrow{AD} \cdot \overrightarrow{AC}$ = ...



(a) 5 (b) 6

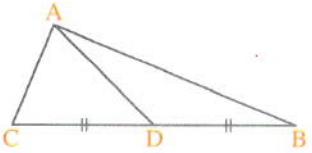
(c) 8 (d) 10

(36) **In the opposite figure :**

ABC is a triangle , D is the midpoint if $\overline{BC}$

If $\| \overrightarrow{AB} + \overrightarrow{AC} \| = \| \overrightarrow{AB} - \overrightarrow{AC} \|$

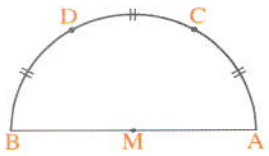
, then m ($\angle$ BAC) = ...



(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

(37) In the opposite figure :



If $\overline{AB}$ is a diameter in circle M , its radius length = 10 cm.

, $m(\overset{\frown}{AC}) = m(\overset{\frown}{CD}) = m(\overset{\frown}{DB})$, then

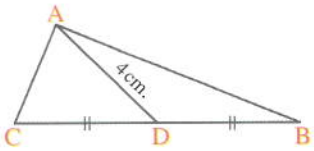
First : $\overrightarrow{CA} \cdot \overrightarrow{CB}$ = ...

(a) Zero (b) 1 (c) 100 (d) 400

Second : $\overrightarrow{BC} \cdot \overrightarrow{BD}$ = ...

(a) Zero (b) 100 (c) 150 (d) 300

(38) In the opposite figure :



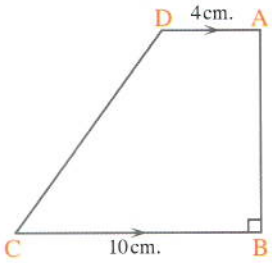
D is the midpoint of $\overline{BC}$, AD = 4 cm.

, then $\overrightarrow{AD} \cdot (\overrightarrow{AB} + \overrightarrow{AC})$ = ...

(a) 16 (b) 24

(c) 28 (d) 32

(39) In the opposite figure :



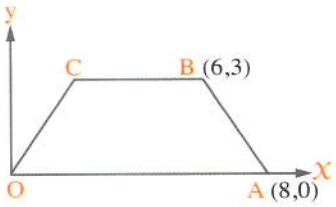
ABCD is a trapezium which a right-angled at B

, $\overline{AD} \parallel \overline{BC}$, then $\overrightarrow{CD} \cdot \overrightarrow{CB}$ = ...

(a) 20 (b) 40

(c) 60 (d) 80

(40) In the opposite figure :



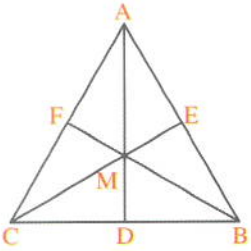
OABC is an isosceles trapezium

, then $\overrightarrow{OB} \cdot \overrightarrow{OC}$ = ...

(a) 16 (b) 18

(c) 21 (d) 24

(41) In the opposite figure :



ABC is an equilateral triangle of side length 4 cm.

, M is the point of intersection of its medians

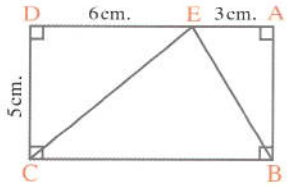
, then $\overrightarrow{MB} \cdot \overrightarrow{CM}$ = ...

(a) $\frac{-8}{3}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{8}{3}$ (d) $\frac{16}{3}$

(42) In the opposite figure :

ABCD is a rectangle , $E \in \overline{AD}$, then $\overrightarrow{EB} \cdot \overrightarrow{EC}$ = ...

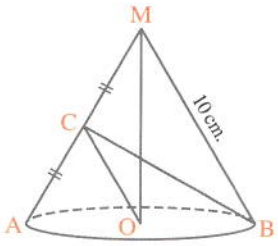


(a) 7 (b) 8

(c) 9 (d) 10

(43) In the opposite figure :

A right-circular cone the circumference of its base = 12π cm. then $\overrightarrow{BC} \cdot \overrightarrow{CO}$ = ...



(a) – 43 (b) – 40

(c) – 37 (d) – 33

20 If $\overrightarrow{A} = (\ell, m, n)$, $\overrightarrow{B} = (\ell`, m`, n`)$ are two unit vectors. Prove that :

$\cos\theta = \ell\ell` + mm` + nn`$ where θ is the measure of the angle between the two vectors.

21 If $\overrightarrow{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$, $\overrightarrow{B} = \hat{i} + k\hat{j} + 5\hat{k}$ find k which makes the measure of the angle between the two vectors $\overrightarrow{A}$, $\overrightarrow{B}$ equals 60°

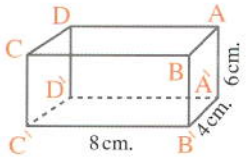
« – 2.12 Approximately »

22 If $\overrightarrow{A} = (2\cos\theta , \log_2 x , \sin\theta)$, $\overrightarrow{B} = (\cos\theta , \log_5 27 , 2\sin\theta)$ and $\overrightarrow{A} \cdot \overrightarrow{B} = 11$ Find the value of x where θ is the measure of the angle between the two vectors $\overrightarrow{A}$, $\overrightarrow{B}$

« 21.04 Approximately »

23 **In the opposite figure :**

ABCD A`B`C`D` is a cuboid , find $\overrightarrow{BD} \cdot \overrightarrow{CA}$



« – 48 »

24 If $\|\overrightarrow{A}\| = \|\overrightarrow{B}\|$ and measure of the angle between $\overrightarrow{A}$ and $\overrightarrow{B}$ is 60° , $\overrightarrow{A} \cdot \overrightarrow{B} = \frac{1}{2}$. Find $\|\overrightarrow{A}\|$ « 1 »

25 If the position vectors of the points A , B , C , D are (1 , 1 , 1) , (2 , 5 , 0) , (3 , 2 , – 3) , (1 , – 6 , – 1) respectively. Find the measure of angle between the two vectors $\overrightarrow{AB}$, $\overrightarrow{CD}$

« 180° »

26 Find the values of k which makes the angle between the two vectors $\overrightarrow{A} = (k , 3k , 1)$, $\overrightarrow{B} = (7 , – 2 , 1)$ obtuse.

« $]-\infty , -1[$ »

27 If $(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 8$, $\|\vec{A}\| = 3\|\vec{B}\|$ **Find :** $\|\vec{A}\|$, $\|\vec{B}\|$ « 3 , 1 »

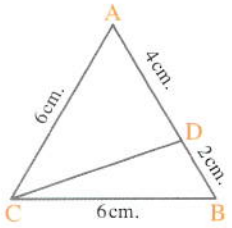
28 If $\vec{A} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{B} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{C} = 3\hat{i} + \hat{j}$ and $(\vec{A} + k\vec{B})$ is perpendicular to the vector $\vec{C}$ Find k « 8 »

29 **In the opposite figure :**

ABC is an equilateral triangle

, $D \in \overline{AB}$ where AD = 2 DB = 4 cm.

Find : $\overrightarrow{CD} \cdot \overrightarrow{CB}$



« 30 »

30 Find the diagonal length of rhombus OBCA whose side length is one unit

, $\sin(\angle AOB) = \frac{24}{25}$, given that AB > OC « 1.2 , 1.6 length unit »

31 The magnitude of vector $\vec{A}$ is 30 , the magnitude of vector $\vec{B}$ is 40 and the measure of the angle between them is 60° **Find :**

(1) $\vec{A} \cdot \vec{B}$ **(2)** $\|\vec{A} + \vec{B}\|$

(3) The component of $\vec{A} + \vec{B}$ in direction of $\vec{A}$ « 600 , $10\sqrt{37}$, 50 »

32 **Find the measures of angles of Δ ABC , where :**

A (2 , – 3 , 4) , B (0 , 1 , 2) , C (– 1 , 2 , 0)

33 ABCD is a rectangle , AB = 40 cm. , BC = 30 cm. Determine the component of vector $\overrightarrow{BD}$ in direction of $\overrightarrow{BC}$, $\overrightarrow{DC}$ « 30 , – 40 »

34 ABCD is a square of side length 10 cm. , determine the scalar product of the two vectors $\overrightarrow{BD}$, $\overrightarrow{BA}$ Calculate the component of vector $\overrightarrow{BD}$ in direction of vector $\overrightarrow{CB}$ « 100 , – 10 »

35 ABCD is a trapezium , $\overline{AD} \parallel \overline{BC}$, $m(\angle A) = m(\angle B) = 90°$ $m(\angle C) = 45°$ AD = 8 cm. , BC = 14 cm. Find the component of $\overrightarrow{CD}$ in direction of $\overrightarrow{CB}$, $\overrightarrow{AD}$ and $\overrightarrow{AB}$ « 6 , – 6 , – 6 »

36 If $\vec{A}$, $\vec{B}$, $\vec{C}$ are unit vectors , such that $\vec{A} + \vec{B} + \vec{C} = \vec{O}$

Find the value of : $\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A}$ « $\frac{-3}{2}$ »

37 If the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ satisfy that :

$\vec{A} + \vec{B} + 2\vec{C} = \vec{O}$ and $\|\vec{A}\| = 1$, $\|\vec{B}\| = 4$, $\|\vec{C}\| = 2$

, find the value of : $m = \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C} + \vec{C} \cdot \vec{A}$ « $\frac{-17}{2}$ »

38 If $\vec{A} + \vec{B} + \vec{C} = \vec{O}$, $\|\vec{A}\| = 3$, $\|\vec{B}\| = 5$, $\|\vec{C}\| = 7$

Prove that measure of angle between $\vec{A}$ and $\vec{B}$ is 60°

39 In Δ ABC , if a , b , c are the side lengths. Prove that :

$\overrightarrow{CA} \cdot \overrightarrow{CB} = \frac{1}{2}(a^2 + b^2 - c^2)$

40 Use vectors to prove that in Δ ABC $(AB)^2 = (AC)^2 + (BC)^2 - 2(AC)(BC)\cos C$

41 Prove by using vectors that the sum of the squares of the diagonal lengths of parallelogram equals the sum of the squares of its side lengths.

42 If D is the midpoint of $\overline{BC}$ in Δ ABC , prove by using vectors :

$(AB)^2 + (AC)^2 = 2(AD)^2 + 2(BD)^2$

43 By using vectors prove that : $\overrightarrow{MA} + \overrightarrow{MB} + \overrightarrow{MC} = \vec{O}$ where M is the centroid of Δ ABC , then prove that : $\overrightarrow{AM} \cdot \overrightarrow{MB} + \overrightarrow{MB} \cdot \overrightarrow{MC} + \overrightarrow{CM} \cdot \overrightarrow{MA} = \frac{1}{2}\left(\|\overrightarrow{MA}\|^2 + \|\overrightarrow{MB}\|^2 + \|\overrightarrow{MC}\|^2\right)$

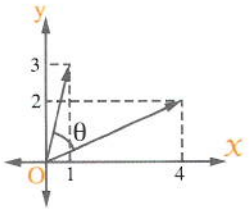
44 ***(1st Session 2018)*** ABC is a triangle in which A (2 , 3 , 1) , B (3 , 5 , 4) , $\overrightarrow{BC} = (-1 , 4 , 0)$, **find :**

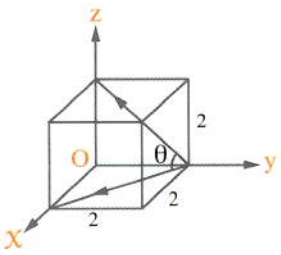
(1) m (∠ ABC)

(2) Vector component of $\overrightarrow{AC}$ in direction of $\overrightarrow{AB}$ « 117° , $\left(\frac{3}{2} , 3 , \frac{9}{2}\right)$ »

45 If $\vec{A} \cdot \vec{B} = -1$, $\vec{A} \cdot \vec{C} = 6$, $\vec{A} \cdot \vec{D} = 5$, $\vec{B} = (3 , 0 , -5)$, $\vec{C} = (2 , 7 , 0)$, $\vec{D} = (1 , 1 , 1)$ Find : $\vec{A}$ « (3 , 0 , 2) »

46 **Find θ in each of the following :**



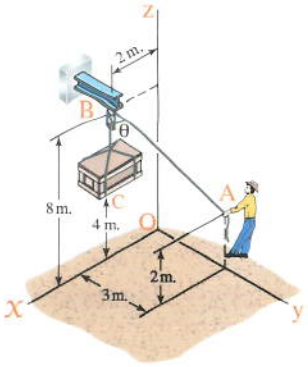


« 45° , 60° »

47 **In the opposite figure :**

If C is the point of intersection of base diagonals of the box.

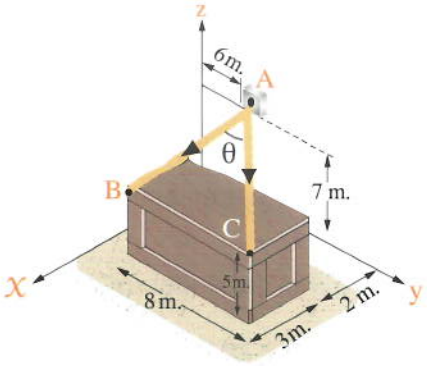
Find the angle between $\overrightarrow{BA}$, $\overrightarrow{BC}$



« 31° »

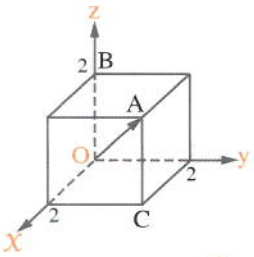
48 **In the opposite figure :**

A wooden box is placed on the floor of a room at one of its coreners 2 m a way from each wall of that corener and tied by two ropes from the vertices B , C to the point A on the wall. Find the measure of angle θ between $\overrightarrow{AB}$, $\overrightarrow{AC}$



« 73° 13̂ »

49 The opposite figure represents a cube with side length 2 length unit, find the component of $\overrightarrow{OA}$ on the vector $\overrightarrow{CB}$



« $\frac{-2}{\sqrt{3}}$ »

50 *(2nd Session 2019)* Find the vector component of $\overrightarrow{AB}$ in the direction of $\overrightarrow{M}$ where A (2 , 1 , 0) , B $\left(3 , 1 , \sqrt{3}\right)$, $\overrightarrow{M} = \left(3 , 2 , 2\sqrt{3}\right)$

Higher skills

51 **Choose the correct answer from those given :**

(1) If $\|\overrightarrow{A}\| = 2$, $\|\overrightarrow{B}\| = 3$, $\|\overrightarrow{C}\| = 12$ and $\overrightarrow{A}$, $\overrightarrow{B}$, $\overrightarrow{C}$ are mutually perpendicular , then $\|\overrightarrow{A} + \overrightarrow{B} + \overrightarrow{C}\| =$ .....

(a) 17 (b) 12 (c) $\sqrt{157}$ (d) 13

(2) If $\vec{A}$ and $\vec{B}$ are two unit vectors and $\| \vec{A} + \vec{B} \| = \sqrt{3}$, then $(3\vec{A} - 4\vec{B}) \cdot (2\vec{A} + 5\vec{B}) =$ .....

(a) $\frac{-17}{2}$ (b) $\frac{-21}{2}$ (c) $\frac{-25}{2}$ (d) -14

(3) If $\| \vec{A} \| = 11$, $\| \vec{B} \| = 23$, $\| \vec{A} - \vec{B} \| = 30$, then $\| \vec{A} + \vec{B} \| =$ .....

(a) 5 (b) 10 (c) 20 (d) 30

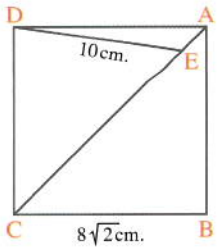
(4) If $\vec{A}$, $\vec{B}$ are unit vectors , then $\| \vec{A} - \vec{B} \|^2 \in$ .....

(a) $[0 , 1]$ (b) $[1 , 4]$ (c) $[0 , 4]$ (d) $[2 , 4]$

(5) In the opposite figure :

If ABCD is a square of side length $8\sqrt{2}$ cm.

, then $\overrightarrow{CB} \cdot \overrightarrow{AE} =$ .....



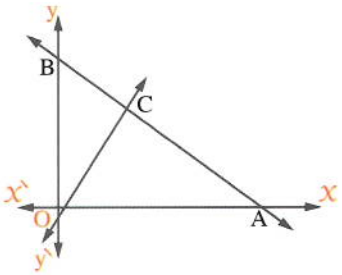
(a) – 24 (b) – 18

(c) – 16 (d) – 12

(6) In the opposite figure :

If the equation of the straight line $\overleftrightarrow{AB}$ is $\frac{x}{12} + \frac{y}{9} = 1$

and AB = 3 BC , then $\overrightarrow{CO} \cdot \overrightarrow{CA} =$ .....



(a) 4 (b) 16

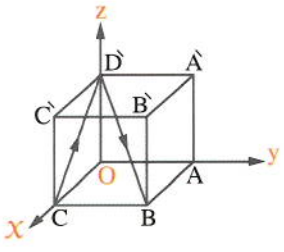
(c) 32 (d) 36

(7) In the opposite figure :

ABCO $\grave{A}\grave{B}\grave{C}\grave{D}$ is a cube of side length

(ℓ) length unit and $\overrightarrow{C\grave{D}} \cdot \overrightarrow{D\grave{B}} = -32$

, then the total area of the cube = ... square unit.



(a) 24 (b) 96

(c) 144 (d) 198

(8) If $\vec{A}$ and $\vec{B}$ are two unit vectors , θ is measure of the angle between $\vec{A}$ and $\vec{B}$, then $\cos \frac{\theta}{2} =$ .....

(a) $\| \vec{A} + \vec{B} \|$ (b) $\frac{1}{2} \| \vec{A} + \vec{B} \|$

(c) $\| \vec{A} - \vec{B} \|$ (d) $\frac{1}{2} \| \vec{A} - \vec{B} \|$

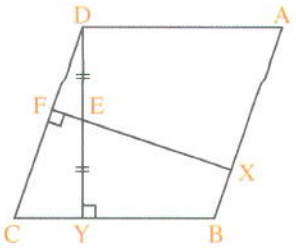
(9) In the opposite figure :

If ABCD is a rhombus

$\overline{DY} \perp \overline{BC}$, $\overline{XF} \perp \overline{CD}$

XE = 3 cm. , EF = 1 cm.

, then $\overrightarrow{CD} \cdot \overrightarrow{CB}$ = ...



(a) $\frac{10}{3}$ (b) 4

(c) 5 (d) $\frac{32}{3}$

52 If $\vec{A} = (2 , -3 , 6)$, $\vec{B} = (-1 , 2 , -2)$ Find the vector $\vec{C}$ where $\vec{C}$ bisects the angle between $\vec{A}$ and $\vec{B}$, $\| \vec{C} \| = 3\sqrt{42}$

53 If $\vec{A} \neq \vec{B} \neq \vec{O}$ Find the necessary condition to make :

(1) $\| \vec{A} + \vec{B} \| = \| \vec{A} \| + \| \vec{B} \|$

(2) $\| \vec{A} + \vec{B} \|^2 = \| \vec{A} \|^2 + \| \vec{B} \|^2$

54 If $\vec{A}$, $\vec{B}$ are non-zero vectors including angle of measure θ. Find the necessary condition to make $\| \vec{A} + \vec{B} \| = \| \vec{A} - \vec{B} \|$